

Pronnoy Dutta

Lead Data Engineer | Senior Data Engineer

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PROFESSIONAL SUMMARY

Databricks-Certified Lead Data Engineer with 6+ years architecting cloud-native data platforms on **AWS and Databricks Lakehouse**, processing **200M+ records daily**. Expert in **PySpark, Spark optimization, Delta Lake, dimensional modeling (SCD Type-2), and data warehousing**, with proven experience leading engineering teams, owning platform architecture, and driving delivery across cross-functional stakeholders. Adept at translating complex business requirements into production-grade architectures that cut runtimes by **30%**, save **\$40K/year** in cloud compute, and build scalable, well-governed data foundations that power enterprise analytics.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PySpark, FastAPI, Java, C#, Bash, Linux

Big Data & Lakehouse: Apache Spark, Databricks, Delta Lake, Medallion Architecture, Hadoop, Hive, YARN

Cloud & Infrastructure: AWS (S3, Glue, EMR, Redshift, Lambda, IAM, DynamoDB), Docker, Kubernetes, Terraform

Data Modeling & Warehousing: Dimensional Modeling, SCD Type-2, Star & Snowflake Schema, Data Warehousing, Data Lakes, OLAP/OLTP

Data Engineering: ETL/ELT, Data Pipelines, NoSQL, CDC, Data Quality, Data Lineage, Schema Governance, Batch & Stream Processing

Leadership & Delivery: Team Leadership, Sprint Planning, Stakeholder Management, Technical Roadmapping, Mentoring, Code Reviews

Orchestration, DevOps & Observability: Apache Airflow, GitHub Actions, CI/CD, Git, Grafana, Prometheus

CERTIFICATIONS

Databricks Certified Data Engineer Professional | AWS Certified Security Specialty | AWS Certified Solutions Architect Associate | Cisco Certified Network Associate (CCNA)

PROFESSIONAL EXPERIENCE

- Axtria** Mar 2022 - Present
Project Lead Apr 2025 - Present | Gurugram, India
- Own end-to-end architecture of cloud-native **data pipelines** on **AWS (S3, Glue, EMR, Redshift)** for a **50M-patient** pharma platform, sustaining **99.9% SLA**; lead **4 engineers** through sprint planning, technical design reviews, and code reviews while owning the technical roadmap for **8+ data products** and serving as primary point of contact for client stakeholders.
 - Led consolidation of fragmented ETL workflows into a standardized, reusable ingestion framework, reducing new data-source onboarding from weeks to days and improving maintainability across **10+ downstream consumers**; drove RCA on **20+ P1/P2 incidents**, cutting repeat failures by **60%** via **Grafana/Prometheus** monitoring.
- Senior Data Engineer* May 2024 - Apr 2025
- Migrated **15+ legacy Hive/SQL pipelines** to distributed **PySpark** on AWS EMR, cutting daily runtime from 9 to 6 hours (**30% gain**), saving **\$40K/year** by resolving data skew, suboptimal partitioning, and oversized shuffles across **200M records/day**.
 - Designed **SCD Type-2 dimensional models** enabling 3+ years of patient-level historical auditability; standardized multi-format ingestion (**JSON, CSV, Parquet, ORC**) via **AWS Glue Data Catalog** with schema-on-read patterns; implemented **data lineage tracking** and schema governance across **10+ upstream source systems**.
- Associate Data Engineer* Mar 2023 - Apr 2024
- Developed production **ETL/ELT pipelines** using **PySpark, AWS Glue, and Control-M** for pharma datasets, achieving **zero-defect delivery** across **12 consecutive releases** via data quality validation (null, referential integrity, row count) cutting downstream defects by **80%**; modeled **star-schema** tables in **Redshift** for **5 analytics use cases**.
- Analyst* Mar 2022 - Mar 2023
- Built a Python-based **Tableau** refresh automation framework across **30+ dashboards**, eliminating **350 hours/year** of manual effort; maintained **ETL/ELT** ingestion and transformation workflows in **PySpark** and **Control-M**, performing data quality validation for pharma commercial datasets.
- Infosys** Nov 2020 - Mar 2022
Systems Engineer Remote
- Architected monthly sales analytics pipelines using **AWS S3, Python, and PySpark** for a retail client across **500+ stores** in 3 regions, cutting reporting effort from 3 days to under 2 hours; built **star-schema data warehouse** with **8 fact/dimension tables** powering executive dashboards.
 - Developed KPI computation logic for **12+ business metrics** (revenue, churn, basket size, store NPS) as single source of truth; optimized complex **SQL** against **100M+ row** transaction tables, reducing report generation time by **40%**.

EDUCATION

B.Tech, Computer Science, Bharati Vidyapeeth College of Engineering, Pune
AWS Community Builders, Official Member, Security Team

Jul 2016 - Jun 2020